

Ibn Tofail University

Rigid body mechanics

Problem Set III

Exercise 1:

Determine in two different ways (directly and using Guldin's theorems) the center of mass of the following solids:

1. A homogeneous semicircle of radius R .
2. A homogeneous quarter disk of radius R .
3. A homogeneous half-disk of radius R .

Correction

Exercise 2:

1. Determine the center of mass of a homogeneous hemisphere (solid half-sphere), of mass m , radius R , and center C .
2. Determine the center of mass of a homogeneous hollow cone, of mass M , height h , and circular base of radius R .
3. Deduce the center of mass of the solid formed by the cone of the second question topped with the hemisphere of the first question.

Correction

Exercise 3:

1. Find the moment of inertia of a homogeneous rod of length $2l$ with respect to an axis that is perpendicular to it and passing through its center of mass.
2. Deduce its moment of inertia with respect to an axis passing through:
 - One of the two ends of the rod.
 - A point on the rod distant $l/2$ from one end.

Correction**Exercise 4:**

Consider a homogeneous disk of radius R and mass M .

1. Directly determine its moment of inertia with respect to its center.
2. Deduce its moment of inertia with respect to one of its diameters.

Correction**Exercise 5:**

A small disk of radius a and center I is removed from a large homogeneous disk of radius R ($R > a$), of center O . The solid obtained is an eccentric circular annulus of mass M . We denote by d the distance between the two centers ($OI = d$).

1. Determine the position of the center of mass of the solid thus obtained.
2. What does this result become if $a = R/2$?
3. Determine the moment of inertia of the solid with respect to the line passing through the two centers. What does this moment of inertia become if $a = R/2$?

Correction**Exercise 6:**

A homogeneous solid S consists of a hollow cylinder, of mass M , radius R , and height h , topped by a solid hemisphere of the same radius R and mass m .

1. Determine the position of the center of mass of solid S .
2. Find the central inertia matrix for each of the two parts of the solid.
3. Deduce the principal inertia matrix of solid S at the center O of the lower base of the cylinder.

Correction